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Relationship between pore size and reversible and irreversible immobilization of ionic liquid electrolytes in porous carbon under applied electric potential

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Transport of electrolytes in nanoporous carbon-based electrodes largely defines the function and performance of energy storage devices. Using molecular dynamics simulation and quasielastic neutron scattering, we investigate the microscopic dynamics of a prototypical ionic liquid electrolyte, [emim][Tf₂N], under applied electric potential in carbon materials with 6.7 nm and 1.5 nm pores. The simulations demonstrate the formation of dense layers of counter-ions near the charged surfaces, which is reversible when the polarity is reversed. In the experiment, the ions immobilized near the surface manifest themselves in the elastic scattering signal. The experimentally observed ion immobilization near the wall is fully reversible as a function of the applied electric potential in the 6.7 nm, but not in the 1.5 nm nanopores. In the latter case, remarkably, the first application of the electric potential leads to apparently irreversible immobilization of cations or anions, depending on the polarity, near the carbon pore walls. This unexpectedly demonstrates that in carbon electrode materials with the small pores, which are optimal for energy storage applications, the polarity of the electrical potential applied for the first time after the introduction of an ionic liquid electrolyte may define the decoration of the small pore walls with ions for prolonged periods of time and possibly for the lifetime of the electrode. *Published by AIP Publishing.*

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Electrical double layer capacitors (EDLCs), which store charge electrostatically at the electrode/electrolyte interface, are promising for high power delivery applications. One of the key challenges for these devices is increasing the energy storage capacity using advanced materials often with unique structural properties. Electrodes in EDLCs are often composed of nanoporous carbon because of its high conductivity, and electrochemical stability over a wide potential range and high surface area. Furthermore, the pore structure can be accurately controlled resulting in variation of the chemical reactivity as well as the unique transport properties of nanoconfined fluids. For example, reports have shown anomalous increases in capacitance when the pore size approaches the diameter of the electrolyte ion.^{1,2} A number of recent simulations have also uncovered unique behaviors of non-aqueous electrolytes in porous carbon electrodes such as oscillation of the capacitance as a function of the pore size.^{3,4} Non-aqueous electrolytes such as room-temperature ionic liquids (RTILs) have received considerable attention because of their negligible vapor pressure, high conductivity, and wide electrochemical window.^{5–7} RTILs are composed of ions at room temperature, and the broad range of possible cation/anion pairs enables the ability to tune the properties of the RTIL for specific applications; in the case of EDLCs based on RTILs, one possible scenario is that by tuning the ion

size(s) to the electrode pore size(s) may result in the optimized supercapacitor performance. Many questions remain about the fundamental aspects of ion diffusion in the nanopores of the carbon electrode, particularly during the charging and discharging process. Moreover, though the transport properties of electrolytes in the porous carbon electrodes under applied potential largely define the function and performance of energy storage devices, many fundamental properties of ion diffusion confined in the nanoporous carbon electrodes^{8,9} are not well understood and require *in situ* techniques coupled with computational modeling.¹⁰

In this work, we explore the microscopic dynamics of a prototypical RTIL, 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (denoted by [emim][Tf₂N]) confined in charged and uncharged carbon electrodes with two distinct pore structures using quasi-elastic neutron scattering (QENS) and molecular dynamics (MD) simulations. An *in situ* electrochemical cell enabled the direct measurement of the RTIL motion within the pores of the carbon electrode as a potential was applied, thus replicating a working EDLC. Two carbon electrodes with a distinct pore structure were synthesized to better understand the effect of pore size on the diffusion processes of the confined RTIL. The first was a soft-templated carbon with hierarchical porosity in which mesopores of 6.7 nm diameter were formed along with smaller micropores

(<2 nm). The second was a purely microporous carbon with slit pores of approximately 1.5 nm.¹¹ QENS measurements indicate nearly bulk-like phase behavior in the larger 6.7 nm pores. In the smaller 1.5 nm pores, slower RTIL diffusion and suppressed phase transitions were observed. MD simulations demonstrate complete reversibility of the EDL in both mesoporous and microporous systems. Since the neutron scattering signal is dominated by the incoherent scattering from hydrogen-bearing cations in the QENS experiment, the ions immobilized near the surface manifest themselves by increased (for immobilized cations) or decreased (for immobilized anions) elastic intensity. For [emim][Tf₂N] in the 6.7 nm pores, the variation of the elastic intensity with the applied potential is indeed reversible, as suggested by the MD results. However, the variation of the elastic intensity for [emim][Tf₂N] in the 1.5 nm pores essentially ceases after the first application of the potential, indicating permanent and irreversible ion immobilization on the carbon surface, which suggests a unique behavior of the RTIL in small micropores with the applied electric potential.

Classical MD simulations were performed to provide guidance about the anticipated effects of varying electrical potential on the structure of ionic liquid EDLs in characteristic mesoporous and microporous carbon systems. Ionic liquid interactions were described using the classical forcefield developed by Lopes *et al.*,^{12,13} which describes ionic liquid molecules in atomistic detail. Partial charges were scaled by a factor of 0.8 to better describe transport properties relative to experimental data.^{12–14} Graphitic carbon atoms were described using parameters from the AMBER forcefield.¹⁵ All simulations were performed with the software package GROMACS 5.0.¹⁶ Bonds were constrained using the LINCS algorithm. The NVT ensemble was used throughout with a timestep of 1 fs. Carbon atoms were frozen in place, but RTIL atoms were integrated with the velocity rescaling thermostat.¹⁷ Non-bonded interactions were computed with a cutoff of 1.1 nm. Electrostatic interactions were computed with the PME method, using a real-space cutoff of 1.1 nm and a Fourier grid spacing of 0.1 nm. Electrical potentials were applied using the fixed charge method (FCM), which, despite its approximations, has been shown to produce results consistent with the more accurate constant potential method (CPM) for low to moderate potentials.^{18,19} In brief, a partial charge is added to each surface carbon atom before the simulation, and the trajectory is analyzed to determine the effective electrical potential. The partial charge is then tuned and the process is repeated until a desired electrical potential is achieved. Two independent simulation cells were constructed to model each porous carbon. To model the 6.7 nm pores, a channel simulation, consisting of two parallel graphene plates separated by 6.7 nm and filled with a bath of RTILs, was constructed. From previous molecular simulation studies, this separation distance is sufficient to treat electrodes independently.^{20,21} To model the microporous carbon, two 1.5 nm slit pores were built and each separated by an 8 nm wide bath of RTILs. For each potential, the following simulation routine was implemented. After a brief energy minimization run and 1 ns of simulation at 300 K, the system was slowly heated to 800 K, run for 4 ns, and slowly cooled down to 300 K and run for 1 ns. Then, data were collected from a 25 ns simulation at 300 K. This process

was first done for a system with no added surface charge to generate an equilibrated initial configuration. To model the presence of an applied electric potential, a partial charge of ± 0.015 or $0.01 e^-$ was added to each electrode surface atom for the mesoporous and microporous systems, respectively. These surface charges correspond to macroscopic charge densities of ± 0.082 and 0.054 C/m^2 , respectively, and have been shown to reproduce potentials near $\pm 2 \text{ V}$.²² Lesser charges were applied to the microporous electrodes to lessen the overscreening effects characteristic of RTILs.^{23,24} Then, to mimic the process of varying these potentials, the partial charges of surface atoms were inverted such that, for example, positively charged surface atoms were subsequently negatively charged.

The results of MD simulations are summarized in Fig. 1. For the mesoporous system, number density profiles are considered across the width of the channel, in which each

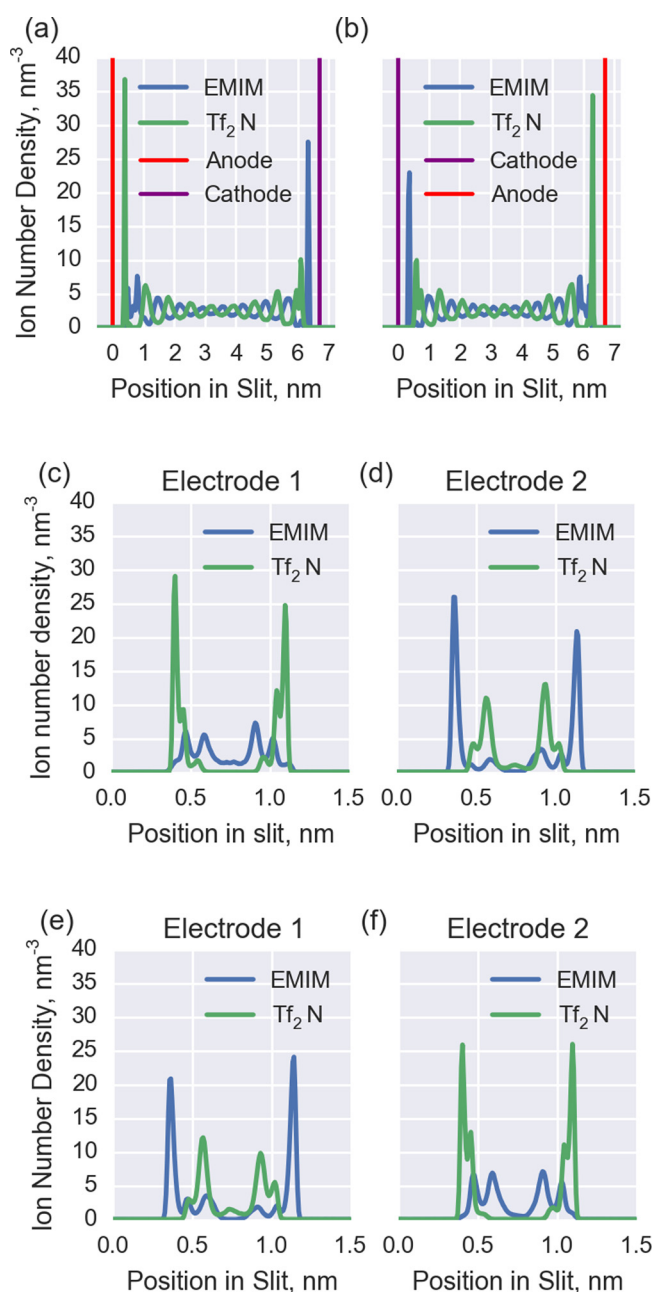


FIG. 1. MD simulation results for the (a) and (b) mesoporous carbon with 6.7 nm pores and (c)–(f) microporous carbon with 1.5 nm pores.

surface is considered to behave independently. Fig. 1(a) shows density profiles of the first charged system. The anode is located on the left, and a large anion peak is found closest to this surface. The opposite electrode and electrolyte peaks are found on the other end of the channel. Both ions display bulk-like behavior in the middle of the pore. Fluctuations are sufficiently small and uniform to be attributed to nanoscale ordering found in ionic liquids.^{25,26} The charges on each surface are then exchanged with each other and resulting number density profiles are shown in Fig. 1(b). In terms of the EDL structure, this is nearly a mirror image of Fig. 1(a) indicating complete reversibility of the EDL in response to varied electrical potentials.

For the microporous system, number density profiles are considered across the width of each pore. Because walls of micropores interact simultaneously with the confined fluid, two pores were analyzed independently. Figs. 1(c) and 1(d) show the number density profiles of ions in the anode and cathode, respectively, of the first charged system. Then, the charges on each electrode are exchanged such that the anode becomes the cathode and vice versa. The resulting density profiles are shown in Figs. 1(e) and 1(f). The new cathode in Fig. 1(e), which was previously the anode, shows an EDL structure nearly identical to the original cathode in Fig. 1(d). Comparison of Figs. 1(c) and 1(f) show a similar result. For both microporous electrodes, a nearly complete reversibility of the EDL structure in response to varied electrical potentials is observed. As noted below, this is not unexpected in the MD simulation due to the short (4 nm) pore lengths and their pristine nature (perfectly parallel surfaces, no defects, no tortuosity in pore geometry, etc.).

To experimentally measure the effect of pore size on the microscopic RTIL dynamics, two types of carbon electrode materials were synthesized (see [supplementary material](#)), soft-templated mesoporous carbon^{27–29} and Matrimid-derived microporous carbon.^{30,31} [emim][Tf₂N], purchased from Io-li-tec, was loaded into the pores of the carbon electrode as described in the [supplementary material](#). The QENS spectra were acquired using a custom electrochemical cell designed specifically for the backscattering instrument at Oak Ridge National Laboratory.³² The electrochemical cell consisted of two aluminum plates with the RTIL-loaded carbon membrane mounted between them. The counter electrode was a high surface area carbon immersed in RTIL in the bottom half of the cell. Only the upper half of the cell was exposed to the neutron beam during the measurement to eliminate scattering from bulk RTIL and from the counter electrode. The size of the upper half of the cell was chosen to fully intercept the incident neutron beam (3 cm by 3 cm) when the cell was perpendicular to the neutron beam. The energy resolution of the spectrometer (width of the elastic line due to neutron scattering by immobile species in the sample) was 3.4 μeV (full width at half maximum). The carbon materials were characterized using both N₂ adsorption and x-ray diffraction (XRD). The pore size distribution, derived from the Kruk-Jaroniec-Sayari (KJS) method,³³ showed a peak at approximately 6.7 nm for the mesoporous carbon while the measurement was unsuccessful for the Matrimid carbon because of the difficulty in measuring these small slit pores. However, techniques such as CO₂

adsorption and x-ray diffraction have been used to quantify the pore size showing that the Matrimid-based carbon has a range of pores from 1 to 2 nm. For MD simulations, we focus on the 1.5 nm pores. XRD and small-angle x-ray scattering (SAXS) data (shown in [supplementary material](#)) are consistent with the presence of micropores in the Matrimid carbon and a hierarchical mesoporous structure in the soft-templated carbon.^{34,35}

The [Tf₂N][−] has no hydrogen atoms, so QENS measurements predominantly probed the single-particle motion of the cation that bears incoherently scattering protons. For characterization of the state of the RTIL in the carbon pores, a temperature scan of the elastic scattering intensity³⁶ was initially performed without the applied potential. The elastic intensities collected on warming up from the baseline temperature and averaged over $0.4 \text{ \AA}^{-1} < Q < 1.4 \text{ \AA}^{-1}$ are presented in Fig. 2. In the larger mesopores, the RTIL exhibited nearly bulk-like phase behavior, showing a melting transition similar to that of the bulk [emim⁺][Tf₂N[−]]. However, this phase transition was suppressed in the 1.5 nm micropores because of nanoconfinement of the RTIL. Accordingly, dynamic QENS runs acquired at 350 K, which were also averaged for $0.4 \text{ \AA}^{-1} < Q < 1.4 \text{ \AA}^{-1}$ (see Fig. 2 inset), show that the RTIL diffuses more freely in the 6.7 nm pores than in the 1.5 nm pores. The QENS scattering signal from the latter sample shows less mobility (smaller wings) compared to the former sample. The suppressed phase transition and mobility are important evidences that the signal from the 1.5 nm pores sample is indeed dominated by the RTIL confined in the micropores.

Because immobilization of the ions on the walls of carbon pores is reflected in the neutron elastic scattering strength,^{37–39} we concentrate on the elastic scattering intensity at zero energy transfer. From Fig. 3, the elastic scattering intensity, but not the quasielastic wings, changed appreciably as a function of the applied potential. This indicates that under applied potential there was no change in the microscopic dynamics of ions diffusing in the center of the pores,

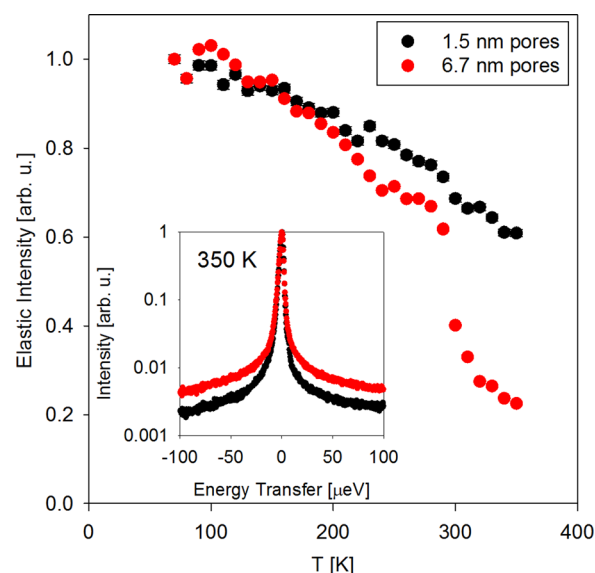


FIG. 2. Diagnostic results of elastic intensity scan and QENS (inset) measurements.

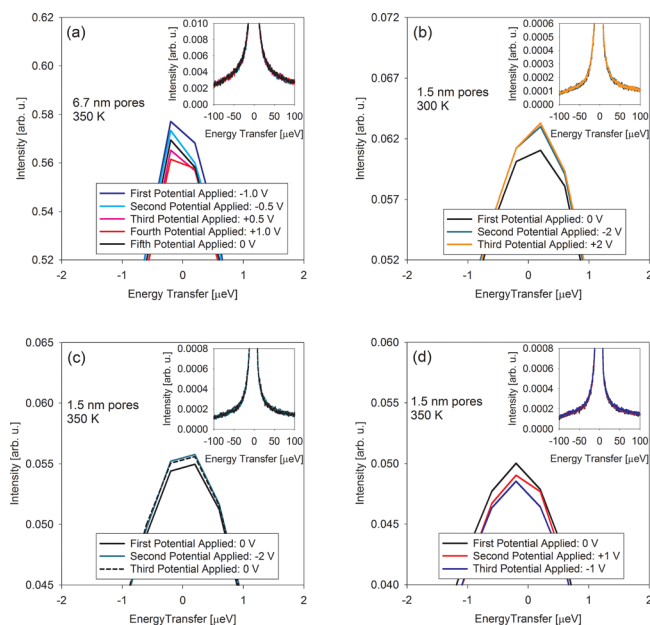


FIG. 3. QENS signal truncated near the elastic peak for (a) mesoporous carbon at 350 K, (b)–(d) microporous carbon at different temperatures and applied voltages. Insets: QENS signal wings.

but substantial change in the dynamics (immobilization) of the ions near the pore walls. The QENS results for the RTIL at 350 K in the 6.7 nm pores are shown in Fig. 3(a). Under a negative potential of -1.0 V, more cations were immobilized near the walls resulting in the increased elastic intensity. Subsequent change of the applied potential to -0.5 V, $+0.5$ V, and $+1.0$ V leads to a monotonic decrease in the elastic intensity. With application of the positive potential, the anions become immobilized near the pore walls resulting in the decreased elastic intensity because neutrons are scattered predominantly by the hydrogen-bearing cations. That is, more immobilized anions are associated with less immobilized cations, hence lower elastic intensity. In full agreement with the MD predictions, the elastic intensity, which is a measure of the amount of ions immobilized near the pore walls, exhibits full reversibility when the applied potential is removed, returning the elastic intensity registered at 0 V to the middle position.

The first measurement for the RTIL in the 1.5 nm pores (Fig. 3(b)) carried out at 300 K yielded an unexpected result; the reversed polarity at $+2$ V did not affect the immobilization of cations near the walls observed after the application of -2 V potential. Because such irreversibility, different from the observation with the 6.7 nm pores, could be attributed to the lower temperature (300 K vs. 350 K), we replaced the sample with another 1.5 nm pore sample and performed the experiment at 350 K. From Fig. 3(c), removal of the potential at the end of the experiment did not restore the original elastic intensity, indicating irreversible immobilization of cations as a result of application of -2 V potential. In order to confirm these two unusual observations, we measured another sample at 350 K (Fig. 3(d)), with a positive potential applied first then application of a negative potential. Another goal of this final experiment was to ensure that the magnitude of the applied potential of 1 V did not exceed that used in the experiment with 6.7 nm pores, in order to

fully validate the comparison between the two pore sizes. Similar to the other measurements involving 1.5 nm pores, the application of $+1$ V followed by -1 V did not reverse the trend in the elastic intensity, indicating irreversible immobilization of anions at a potential of $+1$ V. These three independent measurements for the samples with 1.5 nm pores carried out at different temperatures, applied potentials, and sequence of positive-negative applied potentials clearly indicate a systematic effect of irreversible ion immobilization at the pore walls attained at the first potential application, regardless of the polarity (cation immobilization for the negative and anion immobilization for the positive potential). Because Figs. 3(b)–3(d) display the same intensity span (0.02 in arbitrary units), it is clear that the extent of irreversibility decreases with temperature (from 300 K to 350 K) and increases with maximum applied potential (from 1 V to 2 V). These observations reflect the interplay between the enthalpy term due to ion adsorption on the carbon walls under applied potential and the entropy term that favors ion desorption from the walls. Nevertheless, irreversibility is evident in every measurement taken. Furthermore, since each measurement took about 12 h to ensure adequate neutron counting statistics, the immobilization of ions attained during the initial application of electric potential is long-term and could be considered permanent on the time scale of the experiment of several days.

It is worth briefly discussing some limitations of MD simulations using the slit pore model in view of the reversibility of ion immobilization on the walls of the 1.5 nm pores in the simulation but not in the experiment. A slit pore model was used to isolate the effects of nanoconfinement. Though the simulation trajectories are 25 ns in length, they are considered to characterize equilibrium states, which may or may not be reached in the timescales accessible in the experiment.⁴⁰ For example, the simulation cells included two 4 nm-wide slit pores, of which a short entrance length is disregarded, which produce very short and simple nanopore diffusion pathways relative to the distances and topologies ions are likely to travel in experiment. Also, properties of the highly turbostratic nature of this Matrimid-derived carbon, including such effects as pore size dispersity, surface curvature, and pore accessibility, are not captured in this simple slit pore model.

In conclusion, we have examined the motion of RTIL ions confined within the nanoporous carbon electrodes as a function of applied electric potential. In larger mesopores, QENS indicates an increase in the ion density near the electrode surface with no change in the diffusivity of the RTIL away from the surface. The immobilized ion density changes reversibly as the applied potential is varied. For the smaller micropores (1.5 nm), a similar immobilization of ions at the electrode surface was observed with applied potential. However, for these pores, the immobilization attained at the initial potential application was not reversible on the time scale of experiment duration of several days. The initial potential application leads to irreversible immobilization of cations (for a negative potential) or anions (for a positive potential) which is likely permanent in the technologically important microporous carbon electrodes.

See [supplementary material](#) for description of the porous carbon synthesis, details of the QENS measurement, nitrogen adsorption isotherms, and x-ray scattering data.

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